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zerotier

Zerotier Mesh VPN Service for secure P2P networking between machines

Utility

Usage

```
1 inventory.instances = {  
2   zerotier = {  
3     module = {  
4       name = "zerotier";  
5       input = "clan-core";  
6     };  
7     roles.peer.tags.all = { };  
8     roles.controller.machines.jon = { };  
9     roles.moon.machines.sara.settings.stableEndpoints = [ "77.52.165.46"  
10  };
```



The input should be named according to your flake input. All machines will be peers and connected to the zerotier network. Jon is the controller machine, which will accept other machines into the network. Sara is a moon and sets the `stableEndpoint` setting with a publicly reachable IP, the moon is optional.

Overview

This guide explains how to set up and manage a [ZeroTier VPN](#) for a clan network. Each VPN requires a single controller and can support multiple peers and optional moons for better

connectivity.

Roles

1. Controller

The [Controller](#) manages network membership and is responsible for admitting new peers. When a new node is added to the clan, the controller must be updated to ensure it has the latest member list.

- **Key Points:**
 - Must be online to admit new machines to the VPN.
 - Existing nodes can continue to communicate even when the controller is offline.

2. Moons

[Moons](#) act as relay nodes, providing direct connectivity to peers via their public IP addresses. They enable devices that are not publicly reachable to join the VPN by routing through these nodes.

- **Configuration Notes:**
 - Each moon must define its public IP address.
 - Ensures connectivity for devices behind NAT or restrictive firewalls.

3. Peers

Peers are standard nodes in the VPN. They connect to other peers, moons, and the controller as needed.

- **Purpose:**
 - General role for all machines that are neither controllers nor moons.
 - Ideal for most clan members devices.

Roles

The zerotier service has the following roles:

- controller
- moon
- peer

Options for the `controller` role

`allowedIps`

Extra machines by their zerotier ip that the zerotier controller should accept. These could be external machines.

Type: `list of string`

Default:

[]

Example

```
..  
[ "fd5d:bbe3:cbc5:fe6b:f699:935d:bbe3:cbc5" ]  
..
```

Declared in: clanServices/zerotier/default.nix

Options for the `moon` role

`stableEndpoints`

Make this machine a moon. Other machines can join this moon by adding this moon in their config. It will be reachable under the given stable endpoints.

Type: `list of string`

❖ Example

```
..  
  [ "1.2.3.4" "10.0.0.3/9993" "2001:abcd:abcd::3/9993" ]  
..
```

Declared in: [clanServices/zerotier/default.nix](#)

The `peer` role has no configuration

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